OFFICIAL BOOTH SCRIPT • EXHIBIT 2026

Eloquente Catering System (ECS) with Decision Support System

Authors: John Maverick M. Aquino • Rafael Angelo P. Buscano • John Darev A. Donato • Tyron Jude G. Duran
Adviser: Ms. Geliza Marie I. Alcober • **Industry Partner:** Eloquente Catering Services • **Live Demo:** 4front-ecs.local

PART 1: THE 60-SECOND ELEVATOR PITCH

≈ 1 Minute • For Walkby Evaluators & Casual Visitors

Use this when evaluators stop briefly and ask: "What is your project about?" Deliver with calm confidence, smiling, and pointing to the live website on your screen.

1. THE REAL-WORLD BOTTLENECK

0:00 – 0:15

• EYE CONTACT WITH VISITOR • GESTURE TO EXHIBIT SCREEN

"Good day! In traditional catering businesses like our partner, **Eloquente Catering Services**, booking events is still done manually through paper slips, unverified GCash screenshots, and phone calls. Worse, purchasing ingredients and scheduling banquet staff relies on pure guesswork — causing food waste or sudden service shortages."

2. THE ENGINEERED SOLUTION

0:15 – 0:35

• ACTION: CLICK "ROLES" TAB ON NAVBAR • SHOW 4-STAGE MATRIX

"We engineered **ECS**: an enterprise web platform that automates the entire catering lifecycle across 4 stages. Clients design custom banquets online with instant PayMongo cashless verification, while kitchen staff receive automated prep sheets — completely replacing manual order taking."

3. THE ANALYTICS CORE (DSS LIVE DEMO)

0:35 – 0:50

• ACTION: CLICK "DSS" TAB • DRAG FORECAST SLIDER TO +3 MONTHS

"Our core innovation is a **Decision Support System** running 3 mathematical models: Linear Regression forecasts monthly sales growth with an **R^2 of 0.892** to guide bulk purchasing; Moving Averages predict guest headcount to assign banquet servers; and a 2-Pass Knapsack algorithm automatically builds a 5-course menu matching a client's budget with **&8369;0 overrun risk**."

4. CLOSING & INTERACTIVE INVITATION

0:50 – 1:00

• ACTION: PRESS "H" (OPEN HANDOUT) OR "Q" (SHOW MOBILE QR CODE)

"You can scan this QR code to test the live simulator right on your smartphone, or view our 1-page technical summary. Would you like a quick 3-minute walkthrough of the algorithm?"

Speaker Tip: If the evaluator nods or steps closer, immediately transition to **Part 2 (The 4-Minute Full Walkthrough)** on the next page!

Part 2: The 4-Minute Comprehensive Walkthrough

Structure: Context → 4-Stage Operational Lifecycle → 3 DSS Mathematical Models → Architecture & ISO 25010 Standards

PHASE 1: OPERATIONAL PROBLEM & INDUSTRY PARTNERSHIP

0:00 – 0:45

• ACTION: START ON HERO SECTION • POINT TO ELOQUENTE PARTNER CARD

"Welcome to the ECS capstone exhibit. I am **[Your Name]** from **Team 4FRONT**. Our project is built for **Eloquente Catering Services**, a full-service catering enterprise in Marikina handling over 100 celebrations and banquets every month.

Before ECS, Eloquente suffered from **four critical operational bottlenecks**: first, customer booking required lengthy phone calls and manual price calculations; second, finance clerks had to visually inspect screenshots of GCash and bank receipts, creating fraud risk; third, kitchen staff cooked from handwritten paper slips; and fourth, managers ordered perishable meats and produce based on gut feeling, risking both food spoilage and December peak shortages."

PHASE 2: SEQUENTIAL 4-STAGE CATERING LIFECYCLE

0:45 – 1:45

• ACTION: CLICK "ROLES" ON NAVBAR • CLICK EACH ROLE CARD AS YOU EXPLAIN

"To eliminate these bottlenecks, ECS orchestrates four synchronized operational roles:
• **[Click Stage 1: Client]** Clients customize 5-course banquet menus, receive instant quotations, and lock in bookings with a PayMongo cashless deposit.
• **[Click Stage 2: Sales]** Coordinators manage event calendars with automated date blackout limits, schedule food tastings, and dispatch auto-generated ingredient prep checklists to the kitchen.
• **[Click Stage 3: Finance]** Webhooks automatically verify GCash, Maya, and card payments. Once verified, the double-entry ledger generates BIR-compliant digital receipts with zero manual data entry.
• **[Click Stage 4: Admin]** Executives monitor holiday surcharges, review tamper-proof audit trails recording every action by User ID and IP, and run predictive forecasting models."

PHASE 3: DECISION SUPPORT SYSTEM (DSS) LIVE DEMONSTRATION

1:45 – 3:15

• ACTION: CLICK "DSS" ON NAVBAR • INTERACT WITH THE 3 MODEL TABS LIVE

"Now let's examine the core analytical innovation — our **Decision Support System**, which incorporates three distinct models:

1. Sales Revenue Forecasting: *[Click "Revenue" tab]*
We apply **Simple Linear Regression (SLR): Y = a + b(X)**. As I adjust the horizon slider from +1 to +6 months, the system recalculates projected catering sales based on monthly baseline growth. Notice our empirical validation: an **R^2 of 0.892**, an RMSE of &8369;14,250, and an MAE of &8369;11,800 validated on an 80/20 split. This governs inventory procurement to negotiate supplier volume discounts.

2. Headcount Demand Forecasting: *[Click "Headcount" tab]*
We utilize a **Simple Moving Average (SMA) with seasonal demand smoothing**. Notice that during high-volume December cycles, turnout surges past 2,000 guests. The engine automatically outputs exact banquet staffing ratios at **1 server : 65 guests**, preventing service delays during weddings and corporate galas.

3. Smart Budget Maximizer: *[Click "Budget DSS" tab]*
This is our client-facing optimization engine. When a customer specifies 100 guests and a &8369;50,000 budget, our **2-Pass Knapsack algorithm** first back-calculates 12% VAT and 10% service charge. Pass 1 guarantees full 5-course feasibility (Starter, Main, Side, Dessert, Drink). Pass 2 greedily upgrades dishes to signature tiers within the remaining food budget — achieving **&8369;0 overrun risk**."

Part 3: Architecture & Evaluator Defense Cheat Sheet

Rapid, evidence-backed answers to the most probable questions asked by capstone panellists.

PHASE 4: ARCHITECTURE, STANDARDS & CLOSING

3:15 – 4:00

• ACTION: CLICK "TECH" ON NAVBAR • SHOW ISO BENCHMARKS

"From an engineering perspective, ECS is built on **Laravel 11, Inertia.js, React 18, Tailwind CSS, PostgreSQL**, and the **PayMongo API**. Under **ISO/IEC 25010** software quality evaluation, our test suite achieved 100% functional suitability coverage, zero payment webhook drop rate, and high usability satisfaction (≥ 4.7/5.0).

Thank you for your time, evaluators! We are ready for your questions."

EVALUATOR Q&A CHEAT SHEET (TOP DEFENSE QUESTIONS)

Memorize these bulletproof answers

EVALUATOR QUESTION	BULLETPROOF RESPONSE (SAY THIS PLAINLY)
"Why Simple Linear Regression instead of complex Machine Learning like LSTM or Random Forest?"	**Sample Size & Interpretability:** Catering sales follow clean monthly trendlines governed by seasonal contracts. In small-to-medium enterprise catering, SLR achieved an empirical **R^2 score of 0.892** on an 80/20 chronological split. Complex neural networks require thousands of daily data points and risk overfitting on monthly data, while SLR is mathematically transparent and explainable to business owners.
"How do you prevent payment fraud and fake GCash reference numbers?"	**Server-to-Server Webhook Verification:** Clients do not submit manual transfer slips. Payments route through **PayMongo API**. PayMongo signs every webhook payload with a cryptographic signature header. Our Laravel backend verifies the HMAC signature before marking any invoice as paid, ensuring 100% verified ledger balance.
"What happens if the customer enters an unrealistically low budget in the Maximizer?"	**Feasibility Constraint Check:** The engine calculates the minimum viable 5-course package (&8369;240 × pax × 1.22 taxes/fees). If the budget is below this threshold, the algorithm halts gracefully and outputs the exact shortfall required rather than generating an incomplete menu.
"How does the system benefit the kitchen staff specifically?"	**Automated Kitchen Prep Sheets:** Once a booking is confirmed, ECS aggregates ingredient requirements based on guest headcount and dish recipes, replacing messy handwritten slips with digital, time-stamped culinary prep orders.
"Is this deployed live? Can I test it right now?"	**Yes!** Scan the QR code on our screen or visit **https://jdadonato.github.io/ECS-EXHIBIT/**. The exhibit is a progressive web app (PWA) with full offline caching and responsive controls for smartphones.

Body Language Key: Stand side-by-side with evaluators rather than behind the table. Let them press the sliders or tap the QR code themselves — hands-on interaction creates the highest evaluation scores!